Day 14 — Detection Engineering Consolidation

Name:	Mohammed Farhan
Role:	Associate – Information Security (Intern)
Organization	CyberShelter UAE
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Objective

Consolidate the full Week 2 detection engineering deliverable pack into a single, organized Git repository, update the ATT&CK Navigator coverage layer with all new detections built during Week 2, and run one final end-to-end Atomic Red Team validation confirming the complete purple-team detection pipeline remains reliable ahead of submission.

Part 1 — ATT&CK Navigator Layer Update

The original Day 8 coverage layer (1,233 lines, covering the full ATT&CK technique matrix with a small number of colored/assessed techniques) was reviewed directly rather than recreated from memory, to ensure no existing content was lost or altered.

4 new technique entries added,

Technique	Tactic	Status	Source
T1562.001	Defense Impairment	Green	Day 9 — Defender disabled detection
T1059.004	Execution	Green	Day 12 — Netcat/ncat reverse shell detection
T1547.001	Persistence	Green	Day 11 — Registry Run Key detection
T1021.006	Lateral Movement	Blue	Day 13 — PowerShell lateral movement hunt (log source confirmed, no TP found, detection rule recommended)

2 existing entries updated from unassessed to confirmed coverage:

Technique	Tactic	Status	Source
T1074	Collection	Green	Day 12 — Privileged file staging in /tmp
T1070	Stealth	Green	Day 11 — Clear Windows Event Logs detection

All other techniques and existing color assignments from the original Day 8 layer were preserved unchanged. The updated layer was validated as syntactically correct JSON and confirmed to render correctly in ATT&CK Navigator. The layer file itself (day-14-coverage-map-layer-updated.json) is included in the repository as the primary evidence for this deliverable, since it is designed to be opened and explored interactively rather than summarized in a static image.

Part 2 — Final End-to-End Atomic Red Team Validation

Executed one complete purple-team validation cycle using Atomic Red Team Test T1547.001-1 (Registry Run Key persistence), the most precise and cleanest result from Day 11, selected specifically to serve as a reliable final proof of the full detection pipeline.

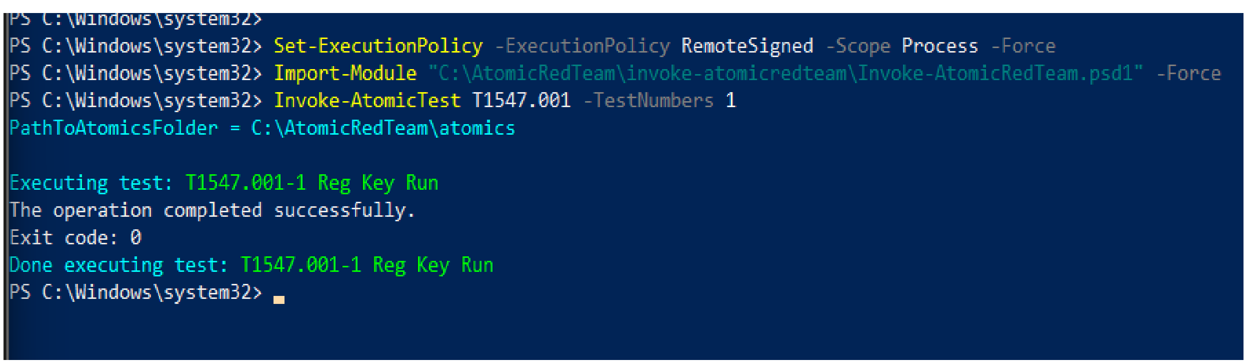
Emulate:

- Ran the atomic test on the Windows VM, adding a registry value under `HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run.`

powershell

- `Invoke-AtomicTest T1547.001 -TestNumbers 1`

Result: Exit code 0, operation completed successfully.



Detect: Sysmon Event ID 13 (Registry value set) captured the action and forwarded it via the Wazuh Agent to the Manager.

Alert: Wazuh rule 92302 fired correctly, generating a level-6 alert. All fields matched the executed test precisely — targetObject showed the exact registry path including the test's distinctive value name ("Atomic Red Team"), and details matched the test's fake executable path exactly.

decoder.name	windows_eventchannel
id	1784116977.993636
input.type	log
location	EventChannel
manager.name	wazuh.manager
rule.description	Registry entry to be executed on next logon was modified using command line application reg.exe
rule.firedtimes	1
rule.groups	sysmon, sysmon_eid13_detections, windows
rule.id	92302
rule.level	6
rule.mail	false
rule.mitre.id	T1547.001
rule.mitre.tactic	Persistence, Privilege Escalation
rule.mitre.technique	Registry Run Keys / Startup Folder

Triage: Confirmed True Positive (within test context) — no ambiguity in the evidence; correctly and precisely tagged with the exact sub-technique T1547.001, not merely the parent technique.

Document: This result is recorded here as confirmation that the detection pipeline — attack execution, Sysmon capture, Wazuh Agent forwarding, and rule-based alerting — functions correctly and reliably end-to-end, immediately prior to Week 2 submission.

Deliverable

Week 2 Deliverable Pack, submitted as a single Git repository containing:

1. **Git repository** with 8 Sigma rules (3 self-written, 5 SigmaHQ community) and 4 custom Wazuh rules (3 detection rules + 1 tuning exception)
2. **ATT&CK Navigator coverage layer**, updated with 6 new/revised Week 2 detections
3. **Alert-tuning report** (Rule 5706 exception, Day 12)
4. **Atomic Red Team coverage table** (5 techniques, Day 11) plus this session's final end-to-end validation
5. **One threat hunt writeup** (Day 13)

Conclusion

The final Atomic Red Team validation confirmed that, after two weeks of iterative rule-building, troubleshooting, and tuning, the core detection pipeline remains accurate and reliable — attack execution, log capture, agent forwarding, and rule-based alerting all functioned correctly in a single clean pass. Updating the ATT&CK Navigator layer to reflect Week 2's genuine progress (moving 6 techniques from unassessed or gap status into confirmed, documented coverage) provides a clear, visual record of the detection engineering work completed this week, ready for review.